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Big Game and Cattle Influence on Aspen Community Regeneration Following Prescribed Fire



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ABSTRACT

Aspen (*Populus tremuloides* Michx.) is a major component of Intermountain forest ecology and relies on periodic disturbance, such as prescribed fire, to perpetuate. On the Manti-LaSal National Forest in central Utah, both big game and cattle depend on forage growing on forested lands, which has contributed to intense conflict. Understanding the effects of browsers on recently burned aspen stands is critical to managing the regeneration of these communities. This study measured the effects of cattle and big game foraging on regenerating aspen communities. Three study sites were selected from a 142-ha prescribed burn conducted in an aspen-conifer stand on the Ferron District of the Manti-LaSal National Forest in 1989. Each of the three study sites was subdivided into four areas and randomly assigned one of the following treatments: big game and cattle exclusion (No Use), big game exclusion (Cattle Use), cattle exclusion (Big Game Use), and open access (Dual Use). Vegetation was sampled in 1991–1994, 1999, and 2005. Density, biomass, height, nested frequency, and cover of aspen suckers were measured. Nested frequency and cover were measured for all other species encountered. Aspen cover, density, and biomass showed a significant year-by-treatment interaction ($P < 0.05$). Aspen and understory regeneration responded similarly to No Use, Cattle Use, and Big Game Use. Dual Use resulted in lower ($P < 0.05$) aspen regeneration and more annual, weedy species in the understory. In 2005, Dual Use aspen cover (4%) was lower ($P < 0.05$) than the other three treatments: Big Game Use (25%), Cattle Use (31%), and No Use (34%). Controlled burning to regenerate aspen will be most successful under light stocking rates for both big game and cattle to allow suckers to develop beyond the browse line (>2 meters).

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Introduction

Aspen (*Populus tremuloides* Michx.), the most widely distributed native North American tree species (Little, 1971; Sargent, 1890), is a major contributor of forest resources in the Intermountain Area (Bartos, 2001). In Utah, Colorado, Nevada, southern Idaho, and western Wyoming, it occupies over 2.02 million ha (Bartos, 1979; DeByle et al., 1989). Aspen communities provide important habitat for a wide variety of wildlife (Chong et al., 2001), summer range for domestic livestock (Bartos et al., 1991; Brown and DeByle, 1989; Gullion, 1977), critical watersheds, wood, recreation resources, aesthetics (Sheppard et al., 2006), and other ecological goods and services (Constanza et al., 1997). These resource-rich communities are also important indicators of ecosystem ecological integrity (Kay, 1997; White et al., 1998).

Aspen reproduce primarily through root suckering stimulated by major disturbances such as burning or clear-cutting (Bartos et al.,

1991; Sheppard et al., 2006). Lack of disturbance and overutilization has interrupted the cycle of aspen regeneration, causing their decline in western forests (Bradley et al., 1992; DeByle et al., 1989; Kay, 1997) and reducing valuable resources (Bartos et al., 1991; Bradley et al., 1992; Canon et al., 1987). In Utah, there has been at least a 60% decrease in aspen since the arrival of Europeans (Bartos, 2001). On the Manti-LaSal National Forest in central Utah, it is estimated that aspen stands decreased 34% between 1915 and 1965 due to conifer invasion (USDA-FS, 1986). This decrease diminished many valuable resources, including understory forage production (DeByle et al., 1989; Julander, 1955; Kleinman, 1973; Stam et al., 2008), negatively affecting the carrying capacity for resident wildlife and livestock.

Prescribed burning is now widely used to promote aspen regrowth and increase productivity of understory species (Bartos and Mueggler, 1981; Brown and DeByle, 1989; Smith et al., 1972). Stand-replacing prescribed fire effectively kills the overstory, warms the blackened soil, damages shallow lateral roots providing hormonal stimulation, releases nutrients, and provides favorable growing conditions for aspen suckers (Sheppard et al., 2006). The high-quality forage in initially low quantities following fire makes the burned areas both highly preferred by livestock and wildlife and highly susceptible to overuse, furthering the decline of aspen (White et al., 1998). In some areas, the animals that

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compete most for these resources are cattle and big game, primarily elk and mule deer, and some moose. Free-ranging beef cattle and big game diets include grass, forbs, and browse in varying proportions depending on selection factors such as site quality and season of use. When forage is plentiful, cattle and big game are generally not in competition for food (Stevens, 1966; Vavra, 1992). However, when highly preferred resources such as postburn aspen stands are scarce, intense competition and damage due to overutilization may result.

Big game/livestock conflicts are a major resource issue on rangelands and have a significant impact on the livestock industry in the 11 western states (USDA-FS, 1992), a situation exacerbated by increases in the number of elk and mule deer (USDA-FS, 2001). Although the damage done by livestock overutilization has long been documented, there is growing evidence that expanding big game populations may be significantly responsible for range overuse and riparian degradation (USDA-FS, 1990). This research is designed to contribute to critical resource information that can aid in the sustainable management of aspen forests. Despite the succession of aspen and understory regeneration following fire being extensively studied, the long-term effects and interactions of cattle and big game foraging on succession are poorly understood. This study was designed to determine the effects of cattle and big game on aspen community regeneration and productivity in burned aspen stands.

Study Area

In the fall of 1989, the Ferron District of the Manti-LaSal National Forest completed a prescribed burn on an aspen-conifer stand located in the 1 863-ha Willow Lake cattle grazing unit of the Ferron C&H Allotment. Roughly average precipitation fell the summer preceding the burn, including 6.5 cm of late July to early August rains. Soil moisture was sufficient to protect the roots of the aspen trees during the burn. The target area was approximately 405 ha, but only 142 ha actually burned, which is approximately 8% of the management unit. The fire created a mosaic of burned and unburned overstory on the North side of Ferron Mountain at the head of Wrigley Creek where the elevation ranges from 2 865 to 2 926 m. The topography of the study area is mountainous with flat meadow areas and undulating shelves that lie along the mountain side. Springs and streams are abundant and provide water for a number of ponds. The high elevation and rain shadow effect of the mountains create a cooler, drier site, resulting in somewhat dwarfed vegetation, including trees. The prefire vegetation consisted of mixed aspen-conifer forest interspersed with mountain big sage (*Artemisia tridentata* Nutt.) and silver sagebrush (*Artemisia cana* Pursh)-grass meadows. The principle conifers were white fir (*Abies concolor* [Gord. & Glend.] Lindl. Ex Hildebr.), Engelmann spruce (*Picea engelmannii* Parry ex Engelm.), subalpine fir (*A. lasiocarpa* [Hook.] Nutt.), and Douglas fir (*Pseudotsuga menziesii* [Mirb.] Franco).

Foraging Pressure

The Ferron C&H Allotment encompasses the burned area and has historically been used by 764 head of cattle from 21 June through 8 October. Following the fire (summer of 1990), no cattle were allowed, but no attempts were made to discourage big game use of the newly burned areas. In 1991, foraging was again permitted to facilitate this study. The study site is located on the Willow Lake Unit of the allotment where the 764 head of cattle (2.5 head per ha) were placed for one month each summer. The month the cattle were in the Willow Lake Unit rotated each year (e.g., September 1991, August 1992, July 1993, September 1994).

Although moose are also found in the area, the big game animals of major concern for this research were primarily elk and, to a lesser extent, mule deer. The study area is located within the Utah Division of Wildlife Resources (UDWR) Manti Elk Herd Unit, which is home to the largest elk herd in Utah. Results from aerial counts of the herd

showed 7 700 in 1991, 9 100 in 1997 (USDA-FS, 2001), 11 100 in 2003, and 9 400 in 2005 (UDWR, 2010). Our study site is located in a critical-use calving and calf-rearing area for some of the elk that winter on Sage Flat, southwest of Ferron, Utah. Approximately 450 wintering elk are on Sage Flat (UDWR, 1990) and one-third (150 head) of them migrate to the Ferron Mountain, Wrigley Hill spring calving range. Observations made in the study area suggest that elk numbers were about 250 head. Each year, elk arrive at the study area just after snow melt in the spring, make use of it sporadically throughout the summer months, and leave around mid-October. Thus there are about 250×4 months = 1 000 elk units each growing season. Although we do not have any estimate on the number of mule deer in the area, they were observed to be present during the study.

Study Sites

Within the burned area, three 0.4-ha locations (study sites) were subjectively chosen to be comparable in exposure, understory vegetation, size, slope, tree density, and burn intensity (Site 1: lat 39°08.222'N, long 111°21.852'W, elevation 2 884 m; Site 2: lat 39°07.779'N, long 111°21.764'W, elevation 2 897 m; Site 3: lat 39° 07.560'N, long 111°20.920'W, elevation 2 921 m). Tree density across all sites averaged 2 427 trees per hectare, or one tree per 4.1 m², of which approximately 25% were aspen and 75% were conifer. All sites were severely burned where all understory vegetation was removed and all overstory trees were killed, leaving standing dead trunks. All three sites had gentle slopes less than 5% but were adjacent to steeper areas greater than 40% that maintained heavy tree overstory cover. The soils of the study sites had a clay loam texture, pH ranging from 6.2 to 7.4, and a color of 10YR 2/1 to 2/2. Precipitation for the sites during the 15 years of the study along with the 30-year average was obtained from the NRCS Buck Flat SNOTEL located at lat 39°08'N, long 111° 26'W (about 6.5 km from the study sites), elevation 2 874 m. Precipitation roughly followed the monthly 30-year average for a total of 80 cm per year during the study, except for slightly below average precipitation during the first 3 years and a spike of 40 cm in 1994 during January to March.

Methods

Each study site was subdivided into four 0.1-ha units, and each unit was randomly assigned one of the four treatments: 1) exclusion of both big game and cattle (No Use), 2) big game exclusion (Cattle Use), 3) cattle exclusion (Big Game Use), and 4) open access to both browsers (Dual Use). Treatments were accomplished by the following: No Use—a permanent 2.1-m tall woven wire fence was maintained year round; Cattle Use—a 2.1-m tall let-down woven wire fence remained up throughout the year except when the cattle came on the management unit; Big Game Use—a 1.2-m tall woven wire fence, which remained down through most of the year to allow free access by big game, was put up to exclude the cattle when they were on the unit; Dual Use—had no fence, allowing open access year round. The Cattle Use treatment technically also allowed big game access during the time that cattle were on the allotment. However, it is unknown how many big game used the Cattle Use areas when cattle were present in the study areas. It is assumed the confounding was reduced because the researchers and UDWR personnel observed that big game tended to avoid allotments occupied by cattle. However, because elk can become habituated to cattle, some of the elk may not have been deterred, resulting in an overestimated impact by cattle. Fences were maintained two or more times per year as needed to prevent any further treatment confounding including during years between data collections.

Vegetation within each treatment was sampled in August of 1991, 1992, 1993, 1994, 1999, and 2005. Data collected consisted of density, biomass, height, cover, and frequency of live aspen suckers. Nested frequency and cover estimations for all other species encountered were also recorded. Using stratified random sampling, five 25-m line

transects were systematically located at even intervals and permanently marked with steel posts within each treatment. There was a 3-m buffer zone next to the fences where no data were collected. Along each transect, five 1-m² quadrats were randomly placed (25 quadrats per treatment). Each quadrat's location was permanently marked by placing metal spikes in the soil in opposite corners of the quadrat frame. Data from the 25 quadrats were averaged into one data point, resulting in 72 experimental units (3 sites × 6 years × 4 treatments).

Density data were collected by counting the number of live aspen suckers within each 1-m² sampling frame. The reference unit method was used to determine aspen forage biomass (Bonham, 1989). A reference unit was obtained by selecting an average-sized aspen shoot to be a unit of measurement to estimate the available forage in each quadrat. Using the reference unit as a visual guide, the estimated number of reference units of forage in each sampling quadrat was recorded. The reference unit was subsequently defoliated, removing all of the current year's growth (leaves and stems), which were then measured for a wet weight and oven dried for 48 hours to obtain a dry weight. Data were then multiplied by the dry weight to obtain the aspen forage biomass estimate for each quadrat. Height data were collected by visually determining and then measuring the average height of all aspen in each quadrat. Cover for each species encountered was determined by estimating the percent of the ground surface covered in the quadrat by the vertical projection of vegetative canopy of that species (Bonham, 1989). The sampling quadrat had seven divisions of increasing size. To obtain nested frequency data, a species occurring in the smallest division received a score of 7 while those occurring in the respectively larger divisions received a correspondingly smaller score (Smith et al., 1986). A species occurring in the largest division received a score of 1. The scores in each plot were then averaged by species and treatment.

Statistical Analysis

Aspen cover, biomass, density, and height were analyzed using SAS Proc Mixed (SAS Inst., 1999) with the study site blocked as a random effect and treatment and year as fixed effects using repeated measures with spatial power structure to account for the unequal spacing between years. Tests for conformance to the assumptions of analysis of variance were performed using SAS Proc Univariate (SAS Inst., 1999), evaluating the Shapiro-Wilk statistic and the necessary transformations made. Cover data, expressed in percentages, were adjusted using the arcsin square-root transformation; biomass and density were adjusted using the square-root transformation; height was normalized using the log 10 transformation; and nested frequency data required no transformation. The Tukey-Kramer procedure was used for mean separation comparing treatments within each year, and letter groupings were determined using an SAS program described by Saxton (1998).

Species richness estimates were calculated from our quadrat data using the jackknife method described by Heltshe and Forrester (1983) comparing the treatments within each year separately. Prevalent species were determined by treatment and year using the method developed by Curtis (1959) in which average frequency was calculated for each species within each treatment and year. Species were then ranked in decreasing order of average frequency. The number of prevalent species presented is equal to the average number of species per quadrat in each treatment block each year.

Results

Aspen

Significant differences were detected in the year by treatment interaction for aspen cover, biomass, and height (Fig. 1). By 2005, Dual Use resulted in lower aspen cover ($P = 0.03$) and height ($P < 0.01$) than No Use and Cattle Use, and lower height ($P < 0.01$) than all other treatments. No differences were detected among No Use, Cattle Use,

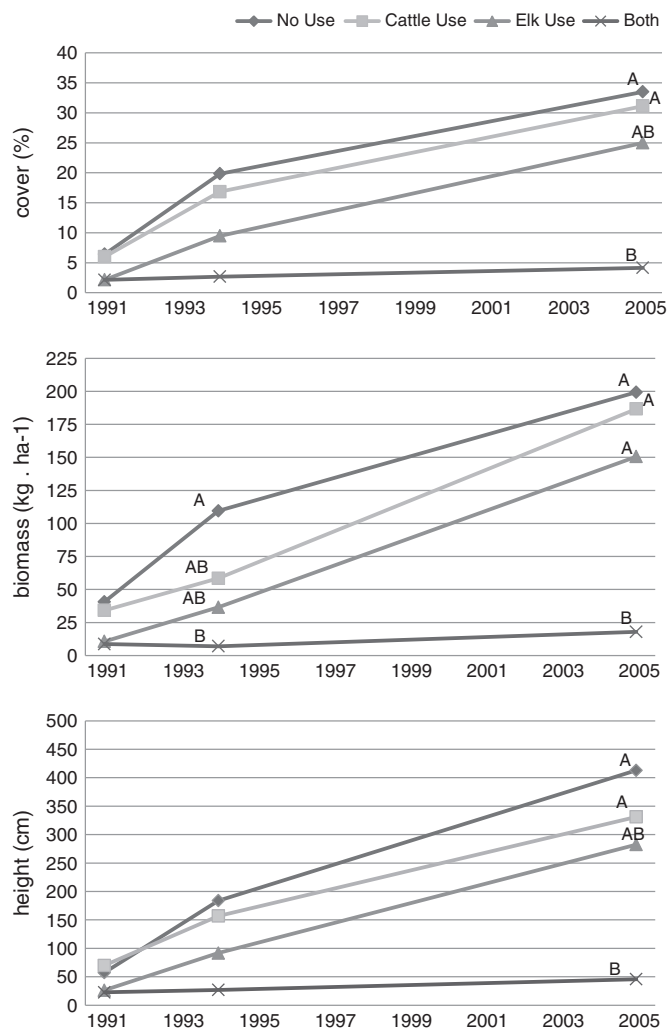


Fig. 1. Year by treatment interaction measuring the effect of four utilization treatments on aspen suckers following a burn in the Ferron district of the Manti-La Sal National Forest over 15 years (1991–2005). Mean separations were performed using $P = 0.05$. Comparisons are presented among treatments within each year; significant differences distinguished by letter designations. Years with no letter designations were not significantly different.

or Elk Use in any year ($P > 0.05$). No differences were detected in aspen cover or height between the Elk Use and Dual Use treatments even in 2005.

Aspen density and frequency did not show a significant year by treatment interaction, but significant differences were detected between years ($P < 0.01$) and between treatments ($P < 0.01$). Aspen density and frequency decreased over time from 38 500 to 10 000 suckers per hectare and a score of 1.6 to 1.0, respectively. Significant decreases in density and frequency were detected only between years 1991–1992 and 1992–1993 ($P < 0.01$). No differences for either measure were detected from 1993–2005. Among treatments, aspen density and frequency were lower ($P < 0.01$) in the Dual Use treatment (8 000 suckers per hectare, a score of 0.7, respectively) than the No Use (27 500, 1.6) and Cattle Use (25 500, 1.6) treatments. No differences in density or frequency were detected ($P > 0.05$) among the No Use, Cattle Use, and Elk Use (16 000, 1.2), and there was no difference between Elk Use and Dual Use ($P > 0.05$).

Understory

Species richness was similar among treatments from 1991–1999. In 2005, richness was lower ($P < 0.05$) in Dual Use (24 species) compared

with the other three treatments (42 species on average). Prevalent species showed the average number of species per plot increase across years, peak in 1999, and then decrease slightly in 2005 (Table 1). In 1991, aspen provided more cover than any other species in all treatments (Table 1). Fifteen years later (in 2005), aspen cover also had the highest cover in No Use, Cattle Use, and Big Game Use treatments. Under Dual Use, Sandberg's bluegrass (*Poa secunda* J Presl) had higher cover than aspen. Two native legumes, Nevada pea (*Lathyrus lanszwertii* Kellogg) and American vetch (*Vicia Americana* Muhl. ex Willd.), had considerable cover by 2005. Both species declined in 2005 under the Dual Use treatment. Dandelion (*Taraxacum officinale* F.H. Wigg.), a well-established, naturalized alien species resistant to foraging, increased in cover under Dual Use utilization.

Discussion

Detrimental impacts of ungulates on regenerating aspen have been well documented (Bartos et al., 1994; Debyle and Winokur, 1985; den Herder et al., 2009; Fairweather et al., 2008; Halofsky et al., 2008; Kay and Bartos, 2000; Mueggler and Bartos, 1977; White et al., 1998). Following the pattern shown by Kay and Bartos (2000), combined use rather than either species alone contributed most to changes in aspen success: Combined use severely stunted regeneration while, after 15 years, cattle only or big game only had no significant effect compared with No Use. As shown by Dockrill et al. (2006), Halofsky et al. (2008), and den Herder et al. (2009), this indicates that some foraging on regenerating aspen can allow for successful recruitment where heavy combined use does not.

Our research did not detect a difference between Cattle Use and Big Game Use. This does not support other research that suggests a greater impact by big game for a variety of reasons. Dockrill et al. (2004) suggests there is benefit for aspen shoots to defer browsing until later in the season when there is more total available forage. In a later study,

Dockrill et al. (2006) indicates that increased resistance to shearing force over the growing season also contributes to the resilience of aspen shoots under late-season browsing. Jones et al. (2009), in a study using simulated browsing to test several gradients of browsing intensities and timing, found that moderate late season browsing, as well as brief early-season-only browsing, can have minimal impacts. Controlled use deferments lessen the impact of cattle compared with uncontrolled timing and length of big game use. Other documented factors contributing to the impact of big game on regenerating aspen stands include browsing terminal buds, rubbing antlers on saplings, and gnawing the bark for phloem (Fairweather et al., 2008). White et al. (1998) suggests big game behavior in the absence of sufficient predators may have additional detrimental effects on aspen beyond mere overpopulation. Although big game may have these effects, our research shows no statistically significant difference between big-game-only and cattle-only use. It should also be noted that the small spatial extent of this study—one burn unit, small plot size, and small sample size—these results are primarily applicable to this area and not all similar areas.

The exclosures in this study were smaller (0.1 ha) than the minimum recommended sizes for accurate elk (4 ha) use (Gross and Knight, 2000), which may have resulted in a lesser impact on the Big Game Use plots. However, as recommended, the cattle fences for the Big Game Use treatment were in place for one month each year when the cattle were on the allotment. The rest of the year, they were completely removed, giving big game free access to the plots. In addition, unlike the pasture setting in the Gross and Knight (2000) paper, where all sides of the exclosures were easily visible to the ungulates, the standing dead trees on our plots visually obscured the size of our exclosures, further minimizing this effect. For these reasons, we believe confounding due to exclosure size was negligible.

It is also possible that mule deer entered the No Use plots because they can jump higher than the 2.1-m fence. However, this appears unlikely because they were never observed in the No Use plots and no evidence of utilization was observed.

The understory species showed a slower response in the first 3 years following fire as evidenced by the low vegetative cover present. Other research has shown more rapid postburn understory growth (Bartos and Mueggler, 1981; Fule et al., 2004; Huisinga et al., 2005). This may have been due to the lower than average precipitation during the first 3 years. Other factors may also include the high level of conifer dominance before the fire, as well as increased fire intensity, both of which decrease the understory seed bank (Lee, 2004; Wang, 2005). In spite of the slow start, understory species cover increased more rapidly by year 9.

As may be expected, foraging-resistant annual forbs and the low-growing Sandberg bluegrass tended to have highest dominance in the Dual Use treatment. The No Use treatment resulted in a greater number of prevalent species and a more even distribution of cover among these species. Cattle Use and Big Game Use fell in between the two and closer to No Use. This supports the same pattern indicated by the aspen data. In contrast to our study, Kay and Bartos (2000) found that cattle use increased introduced species and bare soil. This variation may be explained by a difference in stocking rates or by the prescribed fire in our study, which released a flush of resources not present in the study by Kay and Bartos (2000).

It was visually apparent on and around the study site that the mosaic of the burn created smaller areas of succulent new growth. This may have increased the stocking rate over what it would have been had the burned area been larger or more complete. Similarly, areas that are easily accessible may also be overused while other areas may have inherent utilization restrictions due to topography. Ripple and Larsen (2001) found that jackstraw (areas of thickly fallen logs) significantly protected aspen suckers from browsing. Romme et al. (1995) suggested the lack of aspen regeneration cannot be blamed on any one factor but rather depends on complex interactions among multiple factors.

Table 1

Change in prevalent species with four utilization treatments over a 15-year period following a burn in a quaking aspen stand in the Ferron district of the Manti-La Sal National Forest. The number of prevalent species equals the average number of species per quadrat. The species are listed in order of highest to lowest frequency and % cover is presented. Shaded species are introduced and all others are native

	Species	% Cover			
		No Use	Cattle Use	Big Game Use	Both
1991	quaking aspen	8	8	5	6
	Nevada pea	1	1	0	0
	common dandelion	2	0	0	2
	lambquarters	0	3	0	0
1994	common dandelion	18	11	11	13
	lambquarters	5	3	11	5
	quaking aspen	23	22	15	11
	Nevada pea	4	5	7	4
	American vetch	2	1	2	1
	sedge species	3	1	2	2
	Sandberg bluegrass	2	1	1	3
	common dandelion	9	12	13	15
	common yarrow	4	4	5	3
1999	Sandberg bluegrass	15	12	11	19
	American vetch	5	4	4	5
	Nevada pea	5	7	9	6
	quaking aspen	32	26	23	13
	sedge species	2	1	2	0
	lambquarters	0	1	1	+
	western tansymustard	0	1	1	1
	common dandelion	12	10	7	15
	Sandberg bluegrass	14	13	18	22
	Nevada pea	11	13	16	9
2005	American vetch	13	10	9	3
	common yarrow	2	2	2	3
	quaking aspen	36	34	31	15
	Wood's rose	3	2	0	0
	northern bedstraw	1	+	+	0

Management Implications

Decadent aspen communities that have significant conifer encroachment can respond to burning with vigorous suckering and a renewal of understory species. This research shows that long-term aspen community regeneration following fire was successful and similar to complete protection with exclusion of either cattle or big game. Conversely, combined big game and livestock utilization severely reduced regeneration. Prescribed burning of aspen, with regeneration as an intended goal, may require careful consideration of limiting, but not necessarily eliminating, large ungulate utilization.

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